EMA51005 - Discrete Mathematics

UNIT II - Combinatorics

Final Exam Answer Key

Answers are written in a simple university-exam flow: definition/given, required property, step-by-step working, and conclusion. Strictly following the professor's classwork notes.

Part A - Short Answers

1. Assuming that repetitions are not permitted, how many four-digit numbers can be formed from the six digits 1, 2, 3, 5, 7, 8?

Answer: Total number of digits n = 6, and we form four-digit numbers taking r = 4 digits without repetition.

By the permutation formula: nPr = n! / (n - r)!

6P4 = 6! / (6 - 4)! = 6! / 2! = (1 × 2 × 3 × 4 × 5 × 6) / (1 × 2) = 360.

Hence, 360 four-digit numbers can be formed.

2. Show that if seven colours are used to paint 50 bicycles, at least 8 bicycles will be the same colour.

Answer: Given m = 50 bicycles (pigeons) and n = 7 colours (pigeonholes).

By Generalised Pigeonhole Principle (as in classwork):

Number of bicycles of same colour "e [(m - n)/n] + 1

= [(50 - 7)/7] + 1 = [43/7] + 1 = 6.14 + 1 !' 7 + 1 = 8 (or # (50 - 1)/7# + 1 = 7 + 1 = 8).

Hence, at least 8 bicycles will be the same colour.

3. How many 7 letter words can be formed using the letters of the word BENZENE?

Answer: Total letters n = 7. Letter frequencies: E = 3, N = 2, B = 1, Z = 1.

Number of words = 7! / (3! × 2! × 1! × 1!) = 5040 / (6 × 2) = 5040 / 12 = 420.

Hence, 420 distinct 7-letter words can be formed.

4. State Pigeonhole principle.

Answer: If n + 1 pigeons occupy n holes, then at least one hole has more than one pigeon.

5. Find the minimum number of students needed to guarantee that five of them belong to the same subject having major as English, Maths, Physics, and Chemistry.

Answer: Here n = 4 subjects (holes) and we guarantee k = 5 students in the same subject.

By Generalised Pigeonhole Principle: # (m - 1) / n# + 1 = 5

(m - 1) / 4 = 4 !Ò m - 1 = 16 !Ò m = 17.

Hence, minimum 17 students are needed.

6. Define the generalisation of the pigeonhole principle.

Answer: If m pigeons occupy n holes, then at least one hole has more than [(m - n)/n] + 1 (i.e. at least # (m - 1)/n# + 1) pigeons.

7. A man hiked for 10 hours and covered a total distance of 45 km. It is known that he hiked 6 km in the first hour and only 3 km in the last hour. Show that he must have hiked at least 9 km within a certain period of 2 consecutive hours.

Answer: Total distance = 45 km in 10 hours. Hour 1 = 6 km, Hour 10 = 3 km.

Distance in remaining 8 hours (hours 2 to 9) = 45 - (6 + 3) = 36 km.

Divide these 8 hours into n = 4 disjoint 2-hour periods: (2,3), (4,5), (6,7), (8,9).

By Pigeonhole Principle, at least one 2-hour period covers "e 36 / 4 = 9 km.

8. Define Recurrence relation.

Answer: An equation that expresses the general term of a sequence {a_n} in terms of one or more previous terms of the sequence, namely a_1, a_2, ..., a_{n-1}, for all positive integers n with n "e n_0 (where n_0 is a non-negative integer), is called a recurrence relation for the sequence {a_n}.

9. Write the general solution of recurrence relation.

Answer: For a linear recurrence relation, the general solution is: a_n = a_n^(h) + a_n^(p), where a_n^(h) is the homogeneous solution and a_n^(p) is the particular solution.

For a second-order relation with characteristic roots m_1, m_2:

(i) If m_1 "` m_2 are real and distinct: y_n = A·m_1^n + B·m_2^n.

(ii) If m_1 = m_2 = m are real and equal: y_n = (A + B·n)·m^n.

10. Write the principle of Inclusion and Exclusion formula for the term |A "* B "* C|.

Answer: For any three finite sets A, B, and C:

|A "* B "* C| = |A| + |B| + |C| - |A ") B| - |A ") C| - |B ") C| + |A ") B ") C|.

Part B - 6 Mark Answers

1. Use mathematical induction to prove that n^3 + 2n is divisible by 3, for n "e 1.

Answer: Let P(n) : n^3 + 2n is divisible by 3.
Step 1 (Base Step, n = 1):
P(1) = 1^3 + 2(1) = 1 + 2 = 3, which is divisible by 3. "4 P(1) is true.
Step 2 (Inductive Hypothesis):
Assume P(k) is true for n = k. That is, (k^3 + 2k)/3 = c !Ò k^3 + 2k = 3c for some integer c.
Step 3 (To prove P(k+1) is true):
For n = k + 1: (k + 1)^3 + 2(k + 1) = (k^3 + 3k^2 + 3k + 1) + (2k + 2)
= (k^3 + 2k) + 3k^2 + 3k + 3 = 3c + 3(k^2 + k + 1) = 3[c + k^2 + k + 1],
which is a multiple of 3 and divisible by 3. "4 P(k+1) is true.
Hence, by mathematical induction, n^3 + 2n is divisible by 3 for all n "e 1.

2. Prove by mathematical induction that 1/(1·2) + 1/(2·3) + 1/(3·4) + ... + 1/[n(n+1)] = n/(n+1).

Answer: Let P(n) : 1/(1·2) + 1/(2·3) + ... + 1/[n(n+1)] = n/(n+1).
For n = 1: LHS = 1/(1·2) = 1/2. RHS = 1/(1+1) = 1/2. "4 P(1) is true.
Assume P(k) is true: 1/(1·2) + 1/(2·3) + ... + 1/[k(k+1)] = k/(k+1).
To prove P(k+1) is true: LHS = [1/(1·2) + 1/(2·3) + ... + 1/[k(k+1)]] + 1/[(k+1)(k+2)]
= k/(k+1) + 1/[(k+1)(k+2)] = [k(k+2) + 1] / [(k+1)(k+2)]
= (k^2 + 2k + 1) / [(k+1)(k+2)] = [(k+1)(k+1)] / [(k+1)(k+2)] = (k+1)/(k+2) = RHS.
"4 P(k+1) is true. Hence proved by mathematical induction.

3. A club consisting of 6 men and 7 women, in how many ways can we select a committee of 4 persons which has at least one woman?

Answer: Total men = 6, women = 7. Committee size = 4 with at least 1 woman.
Possible cases (as listed in classwork notes):
1 Woman and 3 Men: 7C1 × 6C3 = 7 × 20 = 140
2 Women and 2 Men: 7C2 × 6C2 = 21 × 15 = 315
3 Women and 1 Man: 7C3 × 6C1 = 35 × 6 = 210
4 Women and 0 Men: 7C4 × 6C0 = 35 × 1 = 35
Total number of ways = 140 + 315 + 210 + 35 = 700.

4. There are 250 students in an engineering college. 188 take Fortran, 100 C, 35 Java, 88 Fortran and C, 23 C and Java, 29 Fortran and Java, and 19 take all three. How many students have not taken any of these three languages?

Answer: Let A, B, C denote students taking Fortran, C, and Java respectively.
Total n = 250. |A| = 188, |B| = 100, |C| = 35.
|A ") B| = 88, |B ") C| = 23, |A ") C| = 29, |A ") B ") C| = 19.
By Principle of Inclusion and Exclusion:
|A "* B "* C| = |A| + |B| + |C| - |A ") B| - |A ") C| - |B ") C| + |A ") B ") C|
= 188 + 100 + 35 - 88 - 29 - 23 + 19 = 323 - 140 + 19 = 202.
Number of students taking none = 250 - |A "* B "* C| = 250 - 202 = 48.

5. Suppose there are 6 boys and 5 girls. (i) In how many ways can they sit in a row? (ii) In how many ways if boys and girls each sit together? (iii) In how many ways if girls sit together and boys do not sit together?

Answer: Total people = 6 + 5 = 11.
(i) Total row arrangements = 11P11 = 11! = 39,916,800.
(ii) Boys sit together (6!) and girls sit together (5!), and the 2 groups can be arranged in 2! ways:
Number of ways = 6! × 5! × 2! = 720 × 120 × 2 = 172,800.
(iii) Girls sit together as 1 block, and boys do not sit together. As derived in classwork:
Treating 5 girls as 1 unit gives 7 units (7! × 5!), minus both groups together (2 × 6! × 5!):
Number of ways = 6! × 5! × 5 = 720 × 120 × 5 = 432,000.

6. Solve the recurrence relation a_{n+2} - 6a_{n+1} + 9a_n = 3·2^n + 7·3^n, n "e 0, given a_0 = 1, a_1 = 4.

Answer: Characteristic equation: m^2 - 6m + 9 = 0 !Ò (m - 3)^2 = 0 !Ò m = 3, 3 (repeated root).
Homogeneous solution: a_n^(h) = (A + Bn)·3^n.
Particular solution:
For 3·2^n: let a_n^(p1) = C·2^n !Ò C·2^{n+2} - 6C·2^{n+1} + 9C·2^n = 3·2^n !Ò C(4 - 12 + 9) = 3 !Ò C = 3.
For 7·3^n: since 3 is a double root, let a_n^(p2) = D·n^2·3^n !Ò substitution gives 18D = 7 !Ò D = 7/18.
General solution: a_n = (A + Bn)·3^n + 3·2^n + (7/18)n^2·3^n.
Using initial conditions a_0 = 1 and a_1 = 4:
n = 0: a_0 = A + 3 = 1 !Ò A = -2.
n = 1: a_1 = (-2 + B)·3 + 6 + 7/6 = 4 !Ò 3B = 4 - 7/6 = 17/6 !Ò B = 17/18.
"4 a_n = (-2 + (17/18)n + (7/18)n^2)·3^n + 3·2^n.

7. How many positive integers n can be formed using the digits 3, 4, 4, 5, 5, 6, 7, if n has to exceed 50,00,000?

Answer: Given 7 digits {3, 4, 4, 5, 5, 6, 7} with repeated digits 4 (twice) and 5 (twice).
To exceed 50,00,000, the first digit must be 5, 6, or 7:
Case 1 (First digit = 5): remaining digits {3, 4, 4, 5, 6, 7} (4 repeated twice): 6! / 2! = 360.
Case 2 (First digit = 6): remaining digits {3, 4, 4, 5, 5, 7} (4 and 5 repeated twice): 6! / (2! × 2!) = 180.
Case 3 (First digit = 7): remaining digits {3, 4, 4, 5, 5, 6} (4 and 5 repeated twice): 6! / (2! × 2!) = 180.
Total positive integers = 360 + 180 + 180 = 720.

8. If 6 people A, B, C, D, E, F are seated about a round table, how many different circular arrangements are possible? If A, B, C are females and others are males, in how many do genders alternate?

Answer: (i) Total circular arrangements of 6 people = (n - 1)! = (6 - 1)! = 5! = 120.
(ii) Gender alternating arrangements: First seat the 3 females around the circle in (3 - 1)! = 2! = 2 ways. Then seat the 3 males in the 3 alternating gaps in 3! = 6 ways.
Number of alternating arrangements = 2! × 3! = 2 × 6 = 12.

9. A team of 11 players is to be chosen from 15 members: (i) One particular player is always included, (ii) Two such players have always to be included.

Answer: (As written in classwork QB 1):
(i) One particular player always included: choose remaining 10 from 14:
(15 - 1)C(11 - 1) = 14C10 = 14! / (4! × 10!) = (11 × 12 × 13 × 14) / (1 × 2 × 3 × 4) = 11 × 13 × 7 = 1001.
(ii) Two such players always included: choose remaining 9 from 13:
(15 - 2)C(11 - 2) = 13C9 = 13! / (4! × 9!) = (10 × 11 × 12 × 13) / (1 × 2 × 3 × 4) = 5 × 11 × 13 = 715.

10. Solve the recurrence relation a_n - 2a_{n-1} = 3^n, a_1 = 5.

Answer: Homogeneous equation: a_n - 2a_{n-1} = 0 !Ò m - 2 = 0 !Ò m = 2. "4 a_n^(h) = A·2^n.
Particular solution: let a_n^(p) = C·3^n !Ò C·3^n - 2C·3^{n-1} = 3^n !Ò 3C - 2C = 3 !Ò C = 3.
"4 a_n^(p) = 3·3^n = 3^{n+1}.
General solution: a_n = A·2^n + 3^{n+1}.
Initial condition a_1 = 5 !Ò A·2^1 + 3^2 = 5 !Ò 2A + 9 = 5 !Ò 2A = -4 !Ò A = -2.
"4 a_n = 3^{n+1} - 2·2^n = 3^{n+1} - 2^{n+1}.

Part C - 10 Mark Answers

1. Solve the recurrence relation a_{n+1} - a_n = 3n^2 - n; n "e 0, a_0 = 3.

Answer: Homogeneous part: a_{n+1} - a_n = 0 !Ò m - 1 = 0 !Ò m = 1. "4 a_n^(h) = A(1)^n = A.
Particular solution: Since m = 1 is a characteristic root, assume a_n^(p) = n(Bn^2 + Cn + D) = Bn^3 + Cn^2 + Dn.
Substitute into a_{n+1} - a_n = 3n^2 - n:
B[(n+1)^3 - n^3] + C[(n+1)^2 - n^2] + D[(n+1) - n] = 3n^2 - n
3Bn^2 + (3B + 2C)n + (B + C + D) = 3n^2 - n.
Comparing coefficients:
3B = 3 !Ò B = 1;
3B + 2C = -1 !Ò 3(1) + 2C = -1 !Ò 2C = -4 !Ò C = -2;
B + C + D = 0 !Ò 1 - 2 + D = 0 !Ò D = 1.
"4 a_n^(p) = n^3 - 2n^2 + n = n(n - 1)^2.
General solution: a_n = A + n^3 - 2n^2 + n. Using a_0 = 3 !Ò A = 3.
"4 a_n = n^3 - 2n^2 + n + 3 = 3 + n(n - 1)^2.

2. Use mathematical induction to prove that n^3 + (n+1)^3 + (n+2)^3 is divisible by 9, for n "e 1.

Answer: Let P(n) : n^3 + (n+1)^3 + (n+2)^3 is divisible by 9.
Step 1 (For n = 1): P(1) = 1^3 + 2^3 + 3^3 = 1 + 8 + 27 = 36 = 9 × 4, divisible by 9. "4 P(1) is true.
Step 2 (Assume P(k) is true): k^3 + (k+1)^3 + (k+2)^3 = 9c !Ò (k+1)^3 + (k+2)^3 = 9c - k^3.
Step 3 (To prove P(k+1) is true): For n = k + 1, expression is (k+1)^3 + (k+2)^3 + (k+3)^3.
= (9c - k^3) + (k^3 + 9k^2 + 27k + 27) = 9c + 9k^2 + 27k + 27 = 9[c + k^2 + 3k + 3],
which is divisible by 9. "4 P(k+1) is true. Hence proved by mathematical induction.

3. Find a formula for the general term a_n of the Fibonacci sequence 0, 1, 1, 2, 3, 5, 8, 13, ...

Answer: Recurrence relation: a_n = a_{n-1} + a_{n-2} !Ò a_n - a_{n-1} - a_{n-2} = 0, with a_0 = 0, a_1 = 1.
Characteristic equation: m^2 - m - 1 = 0 !Ò m = [1 ± " (1 + 4)] / 2 = (1 ± " 5) / 2.
Let m_1 = (1 + " 5)/2, m_2 = (1 - " 5)/2. General solution: a_n = A·m_1^n + B·m_2^n.
Using a_0 = 0: A + B = 0 !Ò B = -A.
Using a_1 = 1: A[(1 + " 5)/2] - A[(1 - " 5)/2] = 1 !Ò A" 5 = 1 !Ò A = 1/" 5, B = -1/" 5.
"4 a_n = (1/" 5) [((1 + " 5)/2)^n - ((1 - " 5)/2)^n] for n "e 0.

4. Prove by mathematical induction that 1^2 + 3^2 + 5^2 + ... + (2n-1)^2 = (1/3)n(2n-1)(2n+1).

Answer: (As written line-by-line in classwork notes):
For n = 1: P(1) = (1/3)(1)(2(1)-1)(2(1)+1) = (1/3)(1)(1)(3) = 1. "4 P(1) is true.
Assume P(k) is true: 1^2 + 3^2 + ... + (2k-1)^2 = [k(2k-1)(2k+1)] / 3.
To prove P(k+1) is true: The next term is (2(k+1)-1)^2 = (2k+1)^2. Add (2k+1)^2 on both sides:
LHS = [k(2k-1)(2k+1)] / 3 + (2k+1)^2
Factor out (2k+1): = (2k+1) [(k(2k-1) + 3(2k+1)) / 3]
= (2k+1) [(2k^2 - k + 6k + 3) / 3] = (2k+1) [(2k^2 + 5k + 3) / 3]
= [(2k+1)(k+1)(2k+3)] / 3 = RHS. "4 P(k+1) is true. Hence proved.

5. Use mathematical induction to show that 1/" 1 + 1/" 2 + 1/" 3 + ... + 1/" n > " n, for n "e 2.

Answer: (As written in 10M classwork notes):
Base step: For n = 2: 1/" 1 + 1/" 2 = 1 + 0.707 = 1.707 > " 2 "H 1.414. (Or P(1) = 1/" 1 = 1). True.
Assume true for k: 1/" 1 + 1/" 2 + ... + 1/" k > " k.
To prove P(k+1) is true:
LHS = (1/" 1 + 1/" 2 + ... + 1/" k) + 1/" (k+1) > " k + 1/" (k+1)
= [" (k)" (k+1) + 1] / " (k+1) > [" (k)" (k) + 1] / " (k+1) = (k + 1) / " (k+1)
= [" (k+1)" (k+1)] / " (k+1) = " (k+1) = RHS.
"4 P(k+1) is true. Hence proved by mathematical induction.

6. Show that a^n - b^n is divisible by (a - b) for all n, n "e 1 using principle of mathematical induction.

Answer: Let P(n) : a^n - b^n is divisible by (a - b).
For n = 1: a^1 - b^1 = a - b, which is divisible by (a - b). "4 P(1) is true.
Assume P(k) is true: (a^k - b^k)/(a - b) = c !Ò a^k - b^k = c(a - b) !Ò a^k = c(a - b) + b^k.
To prove P(k+1) is true: a^{k+1} - b^{k+1} = a^k·a - b^k·b
= [c(a - b) + b^k]a - b^k·b = a·c(a - b) + b^k(a - b) = (a - b)[a·c + b^k],
which is divisible by (a - b). "4 P(k+1) is true. Hence proved by mathematical induction.

7. Find the number of integers between 1 and 250 both inclusive that are not divisible by any of the integers 2, 3, 5 and 7.

Answer: (Verbatim 10M Problem 3 in classwork notes):
Let A_1, A_2, A_3, A_4 be sets of integers divisible by 2, 3, 5, 7 respectively:
|A_1| = # 250/2# = 125, |A_2| = # 250/3# = 83, |A_3| = # 250/5# = 50, |A_4| = # 250/7# = 35.
Pairwise intersections: |A_1 ") A_2| = 41, |A_1 ") A_3| = 25, |A_1 ") A_4| = 17, |A_2 ") A_3| = 16, |A_2 ") A_4| = 11, |A_3 ") A_4| = 7.
Triple intersections: |A_1 ") A_2 ") A_3| = 8, |A_1 ") A_2 ") A_4| = 5, |A_2 ") A_3 ") A_4| = 2, |A_1 ") A_3 ") A_4| = 3.
Four-way intersection: |A_1 ") A_2 ") A_3 ") A_4| = # 250/210# = 1.
By Principle of Inclusion and Exclusion:
|A_1 "* A_2 "* A_3 "* A_4| = (125 + 83 + 50 + 35) - (41 + 25 + 17 + 16 + 11 + 7) + (8 + 5 + 2 + 3) - 1
= 293 - 117 + 18 - 1 = 193.
Number not divisible by 2, 3, 5, 7 = 250 - 193 = 57.

8. How many bit strings of length 10 contain: (a) Exactly four 1's, (b) Atmost four 1's, (c) At least four 1's, (d) An equal number of 0's and 1's?

Answer: (Verbatim 10M Problem 3 in classwork notes):
(a) Exactly four 1's: 10C4 = 10! / (4! × 6!) = 210.
(b) Atmost four 1's: 10C0 + 10C1 + 10C2 + 10C3 + 10C4 = 1 + 10 + 45 + 120 + 210 = 386.
(c) At least four 1's: 10C4 + 10C5 + 10C6 + 10C7 + 10C8 + 10C9 + 10C10
= 210 + 252 + 210 + 120 + 45 + 10 + 1 = 848.
(d) Equal 0's and 1's: 10C5 = 10! / (5! × 5!) = 252.

9. Solve the recurrence relation a_n = 2(a_{n-1} - a_{n-2}), n "e 2 and a_0 = 1, a_1 = 2.

Answer: Equation: a_n - 2a_{n-1} + 2a_{n-2} = 0. Characteristic equation: m^2 - 2m + 2 = 0.
Roots: m = [2 ± " (4 - 8)] / 2 = 1 ± i.
In polar form: 1 ± i = " 2 [cos(< óB' ›w6–âƒÀ/4)].
General solution: a_n = (" 2)^n [c_1 cos(n< óB' ² 5ó" 6–â†ãÀ/4)].
Using a_0 = 1: (" 2)^0 [c_1 cos(0)] = 1 !Ò c_1 = 1.
Using a_1 = 2: " 2 [cos(< óB' ² 5ó" 6–âƒÀ/4)] = 2 !Ò " 2 [1/" 2 + c_2/" 2] = 2 !Ò 1 + c_2 = 2 !Ò c_2 = 1.
"4 a_n = (" 2)^n [cos(n< óB' ² 6–â†ãÀ/4)] (or (1 + i)^{n-1} + (1 - i)^{n-1}).

10. In how many ways can 20 students out of 30 be selected for an activity if: (a) Rama refuses to be selected? (b) Raja insists on being selected? (c) Gopal and Govind insist on being selected? (d) Either Gopal or Govind or both get selected?

Answer: (Verbatim QB 2 in classwork notes): Total students = 30, select = 20.
(a) Rama refuses: choose 20 from remaining 29:
29C20 = 29! / (9! × 20!) = 11 × 23 × 6 × 5 × 13 × 7 × 29 = 10,015,005.
(b) Raja insists: choose 19 from remaining 29:
29C19 = 20,030,010.
(c) Gopal and Govind insist: choose 18 from remaining 28:
28C18 = 13,123,110.
(d) Either Gopal or Govind or both get selected:
Classwork calculation (30C20 - 28C18): 30,045,015 - 13,123,110 = 16,921,905.
(By complement rule 30C20 - 28C20: 30,045,015 - 3,108,105 = 26,936,910).